RSCNet: A Residual Self-Calibrated Network for Hyperspectral Image Change Detection

Liguo Wang, Lifeng Wang, Qunming Wang, and Lorenzo Bruzzone, *Fellow, IEEE*

***Abstract*— Deep learning-based methods (e.g., convolutional neural network (CNN)-based methods) have shown increasing potential in hyperspectral image (HSI) change detection (CD). However, the recent advances in CNN-based methods in HSI CD tasks are mostly devoted to designing more complex architectures or adding additional hand-designed blocks. This increases the number of parameters making model training difficult. In this article, we propose an end-to-end residual self-calibrated network (RSCNet) to increase the accuracy of HSI CD. To fully exploit the spatial information, the proposed RSCNet method adaptively builds interspatial and interspectral dependencies around each spatial location with fewer extra parameters and reduced complexity. Moreover, the introduced self-calibrated convolution (SCConv) helps to generate more discriminative representations by heterogeneously exploiting convolutional filters nested in the convolutional layer. The designed RSC module can explicitly incorporate richer information by introducing response calibration operation. The experiments on four bitemporal HSI datasets demonstrated that the proposed RSCNet method is more accurate than ten widely used benchmark methods.**

***Index Terms*— Change detection (CD), deep learning, hyperspectral images (HSIs), remote sensing, residual network (ResNet), self-calibrated convolution (SCConv).**

I. Introduction

MONITORING changes on the earth surface are important for the sustainable development of human society [1]–[3]. Change detection (CD) is a procedure that recognizes and analyzes variations occurring between multitemporal remote sensing images acquired on the same geographical region [4]. CD has been widely used in various applications, such as urban sprawl analysis [5]–[7], natural disaster assessment [8], ecosystem monitoring [10], [11], and environmental exploration [12]. Compared to other remote sensing images, such as multispectral images (MSIs) and synthetic aperture radar (SAR) images, the hyperspectral images (HSIs) contain rich spectral information and have inherent advantages in detecting subtle land cover changes [1]. With the successful launch of the NASA Earth Observing-1 (EO-1), Italian Space Agency PRISMA, and Chinese Gaofen-5 satellites, the availability of HSIs at the global scale has increased significantly. Therefore, it is of great interest to perform CD based on HSIs, which has received increasing attention in recent years. The main challenges in CD on high-dimensionality HSIs are the redundancy of spectral information, the large computational costs, and the complexity in identifying the channels where changes are detectable.

The algebra-based and transformation-based methods are earliest solutions for CD. For these methods, the feature extraction process usually requires hand-crafted features, which may not effectively extract discriminative information from HSI CD. Moreover, it is difficult to select appropriate thresholds to detect land cover changes by exploiting these features. Recently, an increasing number of deep learning-based methods have been developed. This type of methods proved to be effective to extract high-level abstract features that are more informative than hand-crafted ones and have shown a promising performance in HSI CD. For example, a stacked autoencoder (SAE) was developed to detect changes by learning the spectral features of pixels from the subtraction of bitemporal HSIs [13]. A deep belief network (DBN) composed of a support tensor machine (SVM), and a restricted Boltzmann machine was also adopted to identify changes in HSIs [14]. However, the SAE and DBN methods only consider spectral information of each pixel. These methods ignore spatial features. However, spectral–spatial features, if properly extracted, can improve the performance over spectral features in HSI processing tasks [15].

To take advantage of the spatial information in HSIs, convolutional neural network (CNN)-based methods have been introduced in HSI CD. A CNN trained with self-generated credible labels was adopted to learn the spectral–spatial features of pixels in bitemporal HSIs [16]. A three-direction spectral–spatial CNN (TDSSC) was developed to obtain a change tensor that decomposed along the spectral direction and two spatial directions for HSI CD [17]. A spectral–spatial CNN with Siamese architecture was proposed to extract tensor pairs to detect changes in bitemporal HSIs [18]. Wang *et al.* [19] proposed a general end-to-end 2-D-CNN framework (GETNET), which uses a mixed-affinity matrix that integrates a subpixel representation as the input to detect changes. CNNs have been widely applied in many HSI tasks, such as image classification, super-resolution, target detection, and CD [37], [39].

The recently developed residual network (ResNet) can be considered as an extension of CNNs with skip connections that facilitate the propagation of gradients and perform robustly

Manuscript received November 29, 2021; revised April 7, 2022; accepted May 21, 2022. Date of publication May 23, 2022; date of current version June 9, 2022. This work was supported in part by the National Natural Science Foundation of China under Grant 42171345, Grant 41971297, and Grant 62071084; and in part by Tongji University under Grant 02502350047. *(Corresponding author: Qunming Wang.)*

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Digital Object Identifier 10.1109/TGRS.2022.3177478

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with very deep architectures [20], [21]. Zhong *et al.* [22] designed a spectral–spatial residual network (SSRN) that uses the residual blocks consecutively to learn deep discriminative features from abundant spectral features and spatial contexts for HSI classification. An adaptive multiscale deep fusion residual network (AMDF-ResNet) that includes several residual blocks was proposed to generate multiscale hierarchical features for HSI classification [23]. Jiang *et al.* [24] proposed a novel HSI classification method based on a 3-D separable ResNet (3-D-SRNet) with cross-sensor transfer learning. Xu *et al.* [25] developed an end-to-end model named spectral–spatial residual graph attention network by including spectral residual and graph attention convolution modules for HSI classification. In general, the use of ResNet is mainly concentrated in the classification task for HSI. To the best of our knowledge, ResNet has seldom been considered for CD in bitemporal HSIs. Moreover, the above methods increase computational costs greatly and fail to fully consider the spatial information redundancy, as well as ignore the relationship between neighboring pixels [26].

The attention mechanism inspired by the human visual system can build interdependencies among the spectral and/or the spatial domains. Recently, several attention mechanisms have been introduced to enhance the representation ability of CNNs, such as squeeze-and-excitation network (SENet) [27], convolutional block attention module (CBAM) [28], global context network (GCNet) [29], nonlocal network (NLNet) [30], and dual attention network (DANet) [31]. For HSI processing tasks, a residual attention network, composed of a sequential spectral–spatial attention module and residual blocks, was applied to refine adaptive feature learning in HSI classification [32]. Mdrafi *et al.* [33] adopted attention-based domain adaptation using a ResNet to generate more useful features for HSI classification. Wang *et al.* [34] proposed an SSA-SiamNet method, which integrates CBAM with SiamNet and extracts spectral–spatial-wise features more adaptively from the input patch pairs for HSI CD. However, most of the aforementioned approaches focus on tuning network architectures or adding additional hand-designed blocks to improve CNNs, which leads to complex network structures and massive parameters, coupled with large computational costs.

To alleviate the abovementioned problems, in this article, a residual self-calibrated network (RSCNet) is proposed for the HSI CD. Unlike the aforementioned approaches, the proposed RSCNet method generates more discriminative feature representations by improving the basic convolutional transformation process of CNN (i.e., a process without adjusting network architectures or adding additional hand-designed blocks). Moreover, the proposed RSCNet method introduces fewer extra parameters and reduced complexity. Specifically, compared to the standard convolution that implements all the convolutions over the input in the original space homogeneously, the self-calibrated convolution (i.e., SCConv) separates the inputs into two different scale spaces: an original scale space in which feature maps share the same resolution with the input and a smaller latent space after down-sampling. The embeddings after transformation in the smaller latent space are used as references to guide the feature transformation process in the original feature space, which can adaptively encode informative context in each spatial location, and further, avoid contamination from irrelevant pixels.

The main contributions of this article are as follows.

1) An end-to-end RSCNet framework is proposed to enhance deep feature representations that can be trained from scratch and, furthermore, for more reliable CD.
2) As a baseline, the modified ResNet designed originally for HSI CD is proposed to learn spectral–spatial features from the concatenation of bitemporal HSIs.
3) The introduced SCConv can adaptively calibrate the input feature exploiting different portions of convolutional filters in a heterogeneous way, which builds between-filter communication, and further, makes the feature representation more discriminative.
4) The designed residual self-calibrated convolution (RSC) module composed of SCConv and a residual block can be easily applied to augment residual block with fewer extra parameters. Moreover, the RSC module as a plug and play unit has the potential to be embedded into other baseline networks for different HSI processing tasks.

The remainder of this article is organized as follows. Section II describes the related work of CNN and ResNet. The details of the proposed RSCNet method are illustrated in Section III. The performances of RSCNet are evaluated in the experimental results in Section IV. Further issues regarding RSCNet and open questions for future research are discussed in Section V. Finally, conclusions are presented in Section VI.

II. Related Work

A. Convolutional Neural Networks

In general, CNN is composed of convolutional layers, pooling layers, activation functions, and fully connected layers. The input of CNN should be an image patch $\mathbf{X}$. For a conventional CNN, when $\mathbf{X}$ is operated by the lth convolutional layer with an activation function, the output feature map $\mathbf{H}^{l,k}(\mathbf{S})$ can be obtained as follows:

$$\begin{aligned}\mathbf{H}^{l,k}(\mathbf{S}) &= g\left(\left(\mathbf{W}^{l,k} * \mathbf{H}^{l-1}(\mathbf{S})\right) + \mathbf{B}^{l,k}\right) \\ &= g\left(\sum_{m=1}^{s_l}\sum_{n=1}^{s_l}\sum_{p=1}^{k_{l-1}} \mathbf{W}^{l,k}_{m,n,p} \cdot \mathbf{H}^{l-1}_{i+m-1,j+n-1,p}(\mathbf{S}) + \mathbf{B}^{l,k}\right)\end{aligned} \tag{1}$$

where $\mathbf{H}(\mathbf{S})^{l}_{i,j,p}$ is the output feature map p of the lth layer at pixel $\mathbf{x}(i, j)$. $\mathbf{W}^{l,k}$ and $\mathbf{B}^{l,k}$ denote the kth convolutional kernel and biases of the lth layer, respectively. k_l and s_l are the number of feature maps and the size of the convolutional kernel in the lth layer, respectively. $g(\cdot)$ represents the activation function. For the proposed RSCNet, and the rectified linear function (ReLU) and sigmoid function are used [35].

CNNs have been widely applied in various HSI tasks, such as image fusion, super resolution, target detection, and image classification. Zhang *et al.* [36] proposed a CNN-based reconstruction network with cross-mode information fusion of spatial and spectral modes for HSI and MSI fusion. Wang *et al.* [37] employed dual-channel CNNs, which analyze the information from both single and adjacent bands to

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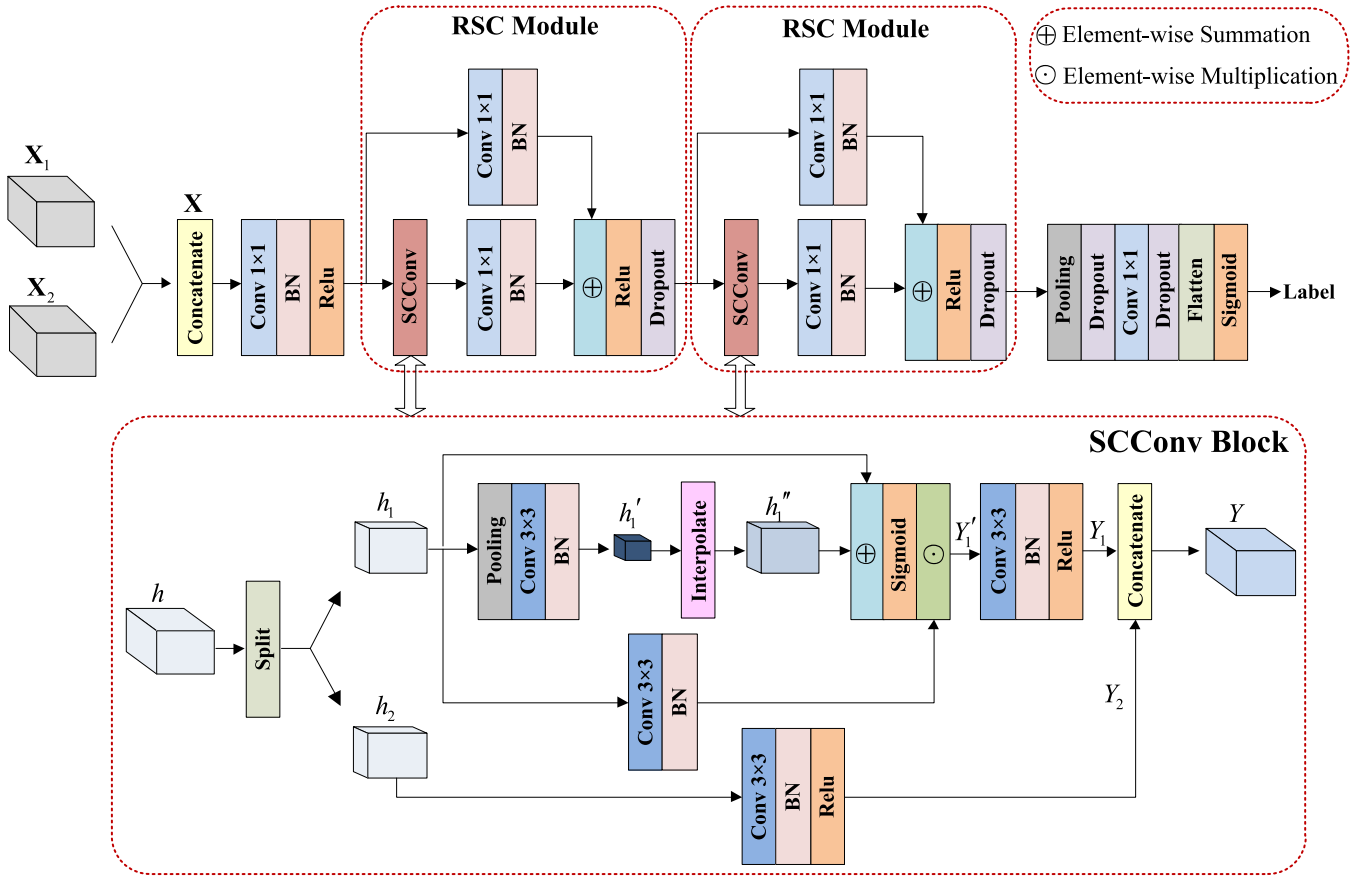


Fig. 1. Overview of the proposed RSCNet for HSI CD.

develop a CNN-based framework for HSI super resolution. Hao *et al.* [38] designed the triplet-pipeline CNN architecture with class-wise loss for HSI classification. The two-stream deep architecture with a class-specific fusion scheme was proposed in [39], which adopts the stacked denoising autoencoders to encode the spectral features and CNN to extract deep spatial features. Furthermore, a number of deeper CNN-based structures have been constructed, such as AlexNet [40], VGG-Net [41], GoogLeNet [42], EfficentNet [43], DenseNet [44], and ResNet [20].

B. Residual Network

ResNet can be regarded as an extension of CNN with shortcut connections, which facilitates the propagation of gradients and performs robustly with a very deep architecture [20]. Therefore, ResNet is used as the baseline in this study. Deep ResNets usually consist of many stacked residual units [20]. Zhao *et al.* [45] integrated enhanced channel and spatial attention units into ResNet (50 layers) for scene classification. In [46], a global–local two-stream architecture combining ResNet (18 layers) with a weakly supervised key area detection strategy was proposed to learn the multiscale representation for scene classification. Li *et al.* [47] proposed a fusion ResNet (50 layers) that can learn the first-order and second-order features simultaneously, and construct the final feature representation by fusing the two types of features. Deng *et al.* [48] developed a high-performance joint framework, which combines ResNet (34 layers) and vision transformer to further boost the discriminative ability of features for image classification.

III. Methodology

An overview of the proposed RSCNet method is shown in Fig. 1. First, a modified ResNet designed specifically for HSI CD is used as a baseline to learn spectral–spatial features from the concatenation of bitemporal HSIs. Second, a novel SCConv is designed to generate more discriminative representations by heterogeneously exploiting different portions of convolutional filters. Third, the designed SCConv is embedded into the residual block to form an RSC module. Finally, the two RSC modules are stacked to extract effective features.

The four main elements of the proposed method, i.e., modified ResNet, SCConv, RSC module, and loss function and optimization, are introduced in Sections III-A–III-D, respectively.

A. Modified ResNet

Bitemporal HSIs are stacked by extending the spectral dimensionality to form the concatenated HSIs. To make full use of the spatial context information, the patches in the concatenated HSI are formed by a center pixel and its surrounding pixels, producing patches of size $s \times s \times 2b$, where s represents

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both the length and width, and b is the number of spectral bands of each HSI.

Let $\mathbf{I} = \{\mathbf{x}(i, j)|1 \leq i \leq h, 1 \leq j \leq w\} \in \mathbb{R}^{h\times w\times b}$ be an HSI, where h and w represent the spatial dimensions. Moreover, the bitemporal HSIs are denoted as I_1 and I_2, respectively. The concatenated HSI I_s can be denoted as follows:

$$I_s = [I_1, I_2] \in \mathbb{R}^{h\times w\times 2b} \tag{2}$$

where the dimensionality of I_s is $2b$. The spectral bands of the pixel at location (i, j) in I_s are represented by $\mathbf{x}(i, j) \in \mathbb{R}^{1\times 1\times 2b}$. The patch centered at $\mathbf{x}(m, n)$ is represented as $\mathbf{X} = \{\mathbf{x}(i, j)|m - (s-1)/2 \leq i \leq m + (s-1)/2, n - (s-1)/2 \leq j \leq n + (s-1)/2\}$, and $\mathbf{X} \in \mathbb{R}^{s\times s\times 2b}$. $\mathbf{X}$ is the input of the proposed RSCNet. For CNN in the proposed RSCNet, the ReLU and sigmoid function are used [35].

In the modified ResNet, two residual units are adopted for the HSI CD task. Moreover, the identity shortcut connection is replaced by a 1×1 convolutional connection [49]. In particular, a dropout layer is added between the residual units to solve the overfitting problem. The residual units in this study can be summarized as follows:

$$h^{l+1} = G\left(\mathcal{F}\left(h^l\right)\right) + f\left(h^l\right)\Big) \tag{3}$$

where h^l and h^{l+1} are the input and output features of the lth unit, respectively. $\mathcal{F}(\cdot)$ is a residual function that represents one or a series of operations, including convolution, pooling, and activation function. $f(\cdot)$ represents the transforms of convolution and batch normalization (BN) operations [50]. $G(\cdot)$ denotes the ReLU activation and dropout operations.

As shown in Fig. 1, a 1×1 convolution with a stride of 1 is adopted in all convolutional layers of the proposed RSCNet except for the SCConv block part, and kernel numbers are all set to K.

The network ends with a fully connected layer containing two filters and a sigmoid activation function. The output of the proposed network is a binary label that indicates whether the pixel has changed or not.

B. SCConv

In grouped convolutions, the features are usually extracted from multiple parallel branches with similar structures independently. Then, the outputs from each branch are concatenated as the final output [26]. Inspired by this operation, the designed SCConv also adopts the split-transformations-merge strategy. The learnable convolutional filters are split into multiple portions in the proposed SCConv. However, each portion of the filters is not equally treated but is in charge of an appropriate functionality.

The workflow of the proposed design is illustrated in Fig. 1. The input of the SCConv h is uniformly split into two portions $\{h_1, h_2\}$. Then, each portion is sent into a pathway for extracting different types of contextual information. In the first pathway, the input feature h_1 is adaptively calibrated by heterogeneously exploiting different portions of convolutional filters, which builds between-filter communication. In the second pathway, we perform a simple convolution operation on h_2, which aims to retain the original spatial context. Thus, the concatenated output Y fuses information from self-calibrated and original spatial contexts, which can generate more discriminative representations.

In the first pathway, the input h_1 is analyzed by an SCConv that provides as output Y_1. It starts from the process as follows. The intermediate feature of the small-scale space h_1' is defined by

$$h_1' = f(\text{Pool}(h_1)) \tag{4}$$

where $\text{Pool}(\cdot)$ represents the pooling operation. Through the experimental analysis presented in Section IV-E4, the average pooling with filter size $r \times r$ and stride r is adopted, and r is set to 3 in this study.

h_1' is scaled to the original feature space using an up-sampling strategy, described as

$$h_1'' = \text{Up}\left(h_1'\right) \tag{5}$$

where $\text{Up}(\cdot)$ is a interpolation operator. Three interpolation strategies, namely, bilinear, nearest neighbor, and bicubic, will be discussed in Section IV-E3. The calibration operation is formulated as follows:

$$Y_1' = f(h_1) \cdot \sigma\left(h_1 + h_1''\right) \tag{6}$$

where $\sigma(\cdot)$ and "$\cdot$" denote the sigmoid function and the element-wise multiplication, respectively. Then, the output of the first pathway Y_1 is calculated as

$$Y_1 = g\left(f\left(Y_1'\right)\right). \tag{7}$$

For the second pathway, the output Y_2 is calculated as

$$Y_2 = g(f(h_2)). \tag{8}$$

Finally, both outputs of the two pathways, that is, Y_1 and Y_2, are concatenated together as the output Y

$$Y = [Y_1, Y_2] \in \mathbb{R}^{s\times s\times K}. \tag{9}$$

From Fig. 1, a 3×3 convolution with a stride of 1 is adopted in all convolutional layers of the SCConv blocks, and the kernel numbers are all set to $K/2$.

C. RSC Module

Considering the superior performance of the residual block in ResNet, the SCConv block is embedded into the residual block, as shown in Fig. 1. The RSC module is composed of an SCConv block, two convolutional layers, two BNs, a shortcut connection, a ReLU activation function, and a dropout layer. The arrangement of these basic layers or blocks is shown in Fig. 1, which can be formulated as follows:

$$h_{sc} = G(f(\mathcal{F}_{sc}(h)) + f(h)) \tag{10}$$

where $\mathcal{F}_{sc}(\cdot)$ represents the SCConv operation. As shown in Fig.1, two RSC modules are stacked in the proposed RSCNet.

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TABLE I
NUMBERS OF PIXEL PAIRS IN TRAINING AND TEST SETS FOR THE FOUR CONSIDERED DATASETS

Dataset	Training Set			Test Set		
	Unchanged	Changed	Total	Unchanged	Changed	Total
Farmland	914	2236	3150	17363	42487	59850
River	5094	485	5579	96791	9213	106004
Hermiston	2984	715	3699	56704	13584	70288
Bay Area	342	392	734	33869	38878	72747

D. Loss Function and Optimization

An appropriate loss function is necessary to optimize the designed network during model training. Binary cross entropy is commonly applied to binary classification tasks. Therefore, binary cross entropy is used as the loss function of the proposed RSCNet for HSI CD. The loss function for binary cross entropy is expressed as follows:

$$\text{Loss}_{\text{BCE}} = -\frac{1}{M}\sum_{i=1}^{M}\left[y_i \times \log\left(\hat{y}_i\right) + (1-y_i) \times \log\left(1-\hat{y}_i\right)\right] \tag{11}$$

where y_i and $\hat{y}_i$ represent the true label and the predicted proability of the ith pixel pair, respectively. Moreover, the labels of the unchanged and changed pixel pairs are set to $y_i = 0$ and $y_i = 1$, respectively. M is the number of training sample pairs in a mini-batch. The parameters in the model are updated using backpropagation and stochastic gradient descent.

IV. EXPERIMENTS

A. Description of Datasets

Four HSI datasets were used to demonstrate the performance of the proposed RSCNet method.

In this study, the first three HSI datasets (i.e., Farmland, River, and Hermiston datasets) were collected by the Hyperion sensor onboard the EO-1 satellite. The EO-1 Hyperion sensor covers wavelengths ranging from 0.4 to 2.5 μm with 242 spectral bands. It provides HSIs with a spatial resolution of approximately 30 m and a spectral resolution of approximately 10 nm. The Bay Area dataset was acquired by the Airborne Visible InfraRed Imaging Spectrometer (AVIRIS) from the NASA's Jet Propulsion Laboratory (JPL). AVIRIS covers the spectral radiance in the wavelength ranging from 0.4 to 2.5 μm with 224 spectral bands. In general, the HSIs acquired by AVIRIS have a spatial resolution of 20 m and a spectral resolution of 10 nm.

The first dataset, named "Farmland" [19], covers a farmland in Yancheng, Jiangsu province, China. The bitemporal HSIs were acquired on May 3, 2006, and April 23, 2007. The spatial size is 450 $\times$ 140 pixels, and 155 bands were considered after removing the low signal-to-noise ratio spectral bands. The false-color composites (bands 33, 22, and 11 as RGB) of the two images are shown in Fig. 2(a) and (e). The changes in the dataset were mainly caused due to crop rotation.

The second dataset, named River [19], encompasses a river region in Jiangsu province. The false-color composites (bands 33, 22, and 11 as RGB) of the bitemporal images are shown in Fig. 2(b) and (f). The bitemporal HSIs were acquired on May 3, 2013, and December 31, 2013. The spatial extent was 463 $\times$ 241 pixels, and 198 bands were selected for CD after removal of noisy bands. The changes in the dataset are mainly caused by the removal of sediment from the river.

The third dataset, named Hermiston, covers an irrigated agricultural field in Hermiston City, Umatilla County, OR, USA. The two HSIs were acquired on May 1, 2004, and May 8, 2007. This dataset has a spatial size of 307 $\times$ 241 pixels and contains 154 bands after removing noisy bands. The bitemporal images are shown in Fig. 2(c) and (g) (bands 33, 22, and 11 as RGB). The changes in the dataset are caused mainly by regulation of irrigation areas.

The fourth dataset, named Bay Area, covers an area in the San Francisco Bay Area, CA, USA. The two HSIs were acquired in 2013 and 2015 and are shown in Fig. 2(d) and (h) (bands 33, 22, and 11 as RGB). The spatial size is 600 $\times$ 500 pixels, and 224 spectral bands were considered.

B. Experimental Setup

For testing, all patch pairs were divided into training and test sets according to a predefined proportion. For a fair comparison, 5% of the samples from the Farmland, River, and Hermiston datasets and 1% of the samples from the ground references of the Bay Area dataset were randomly selected as the training sets. The details are presented in Table I. To increase the learning ability, all input data from the four datasets were standardized to zero mean values with unit variance.

The experiments were implemented using a computer with an NVIDIA GeForce GPU RTX3090 with 24-GB RAM. To avoid biased estimation, ten independent tests were carried out using the TensorFlow and Keras.

The RSCNet was trained from scratch. The optimizer was the adaptive moment estimation (Adam) with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. Moreover, the learning rate was set to 0.001 in the first 100 epochs and 0.0001 in the second 100 epochs, and early stopping was used to avoid overfitting.

To quantitatively evaluate the proposed method, we used the overall accuracy (OA), kappa coefficient (Kappa), precision (Pr), recall (Re), and F1-score (F1) as assessment metrics [34].

C. Parameter Setting

In the proposed RSCNet method, several hyperparameters, such as the kernel numbers of convolutional filters, batch size, patch size, and the training sample proportion, can affect the model training progress and, finally, the detection

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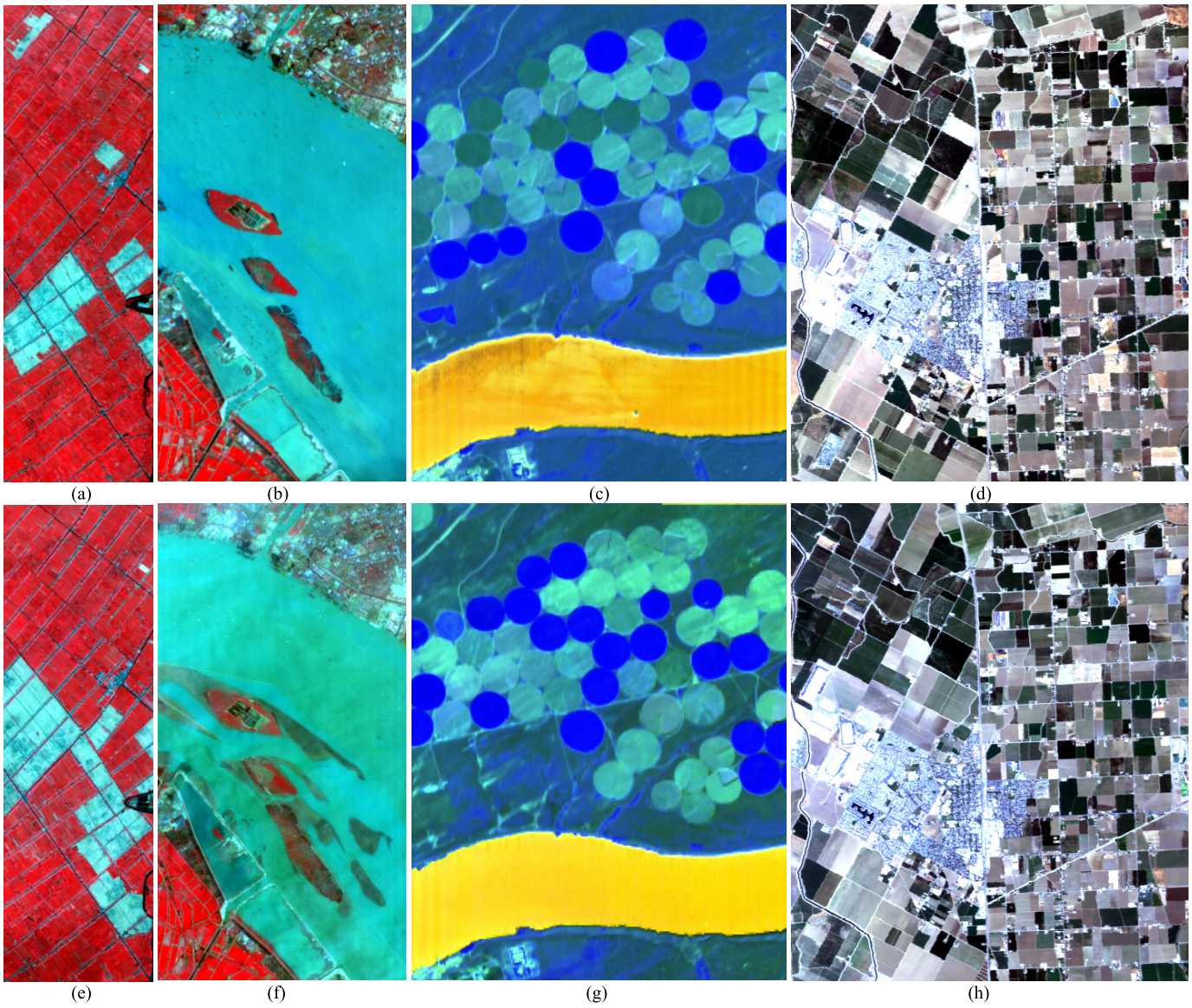


Fig. 2. Four HSI datasets used in the experiments (bands 33, 22, and 11 as RGB): (a) and (e) are Farmland images acquired on May 3, 2006 and April 23, 2007; (b) and (f) are River images acquired on May 3, 2013 and December 31, 2013; (c) and (g) are Hermiston images in 2013 and 2014; and (d) and (h) are Bay Area images in 2013 and 2015.

performance. Thus, the impact of these hyperparameters was explored. Specifically, when the influence of a certain parameter was analyzed, the other parameters were fixed in the experiments.

1) Impact of the Kernel Numbers: The kernel numbers of the convolutional filters affect the computational cost and representation capacity of the RSCNet. The same kernel number is used in each convolutional layer of the RSCNet. A set of kernel numbers (4, 8, 16, 24, and 32) was examined in preliminary experiments. As shown in Fig. 3(a), as the kernel numbers increase from 4 to 32, the OA value also increases for the River and Hermiston datasets. The optimal kernel number is 16 for the Bay Area dataset and 32 for the other three datasets.

2) Impact of Batch Size: To explore the effect of patch size on the performance of RSCNet, different batch sizes (i.e., 16, 32, 64, 128, 256, and 512) were examined. From Fig. 3(b), one can see that by the increasing in the batch size, the OA value decreases for the Bay Area dataset. As the kernel numbers increase, the OA value first increases and then decreases for the Farmland and River datasets. The optimal batch sizes for the Farmland, River, Hermiston, and Bay Area datasets are 32, 32, 16, and 64, respectively.

3) Impact of Patch Size: The size of the input patch reflects the amount of spatial information considered around the center pixel. To evaluate the influence of different patch sizes on the proposed method, we considered a set of four patch sizes: 3×3, 5×5, 7×7, and 9×9. As shown in Fig. 3(c), by increasing the patch size, the OA decreases gradually for the Farmland, River, and Hermiston datasets. The patch size of 5×5 leads to the largest OA for the Bay Area dataset. The optimal patch size is 3×3 for the other three datasets.

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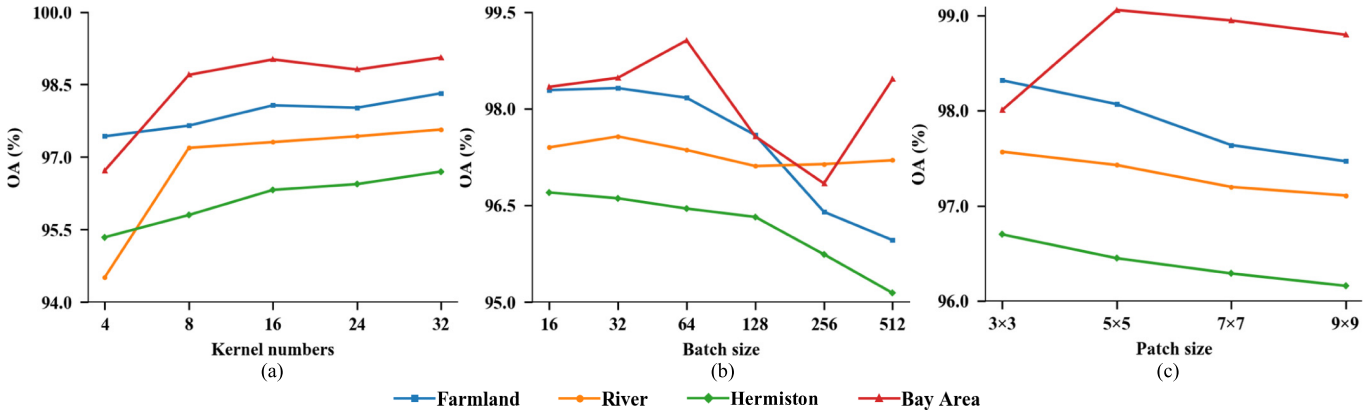


Fig. 3. Accuracy (in terms of OA) of RSCNet with different hyperparameter values for the four datasets. (a) Kernel numbers. (b) Batch size. (c) Patch size.

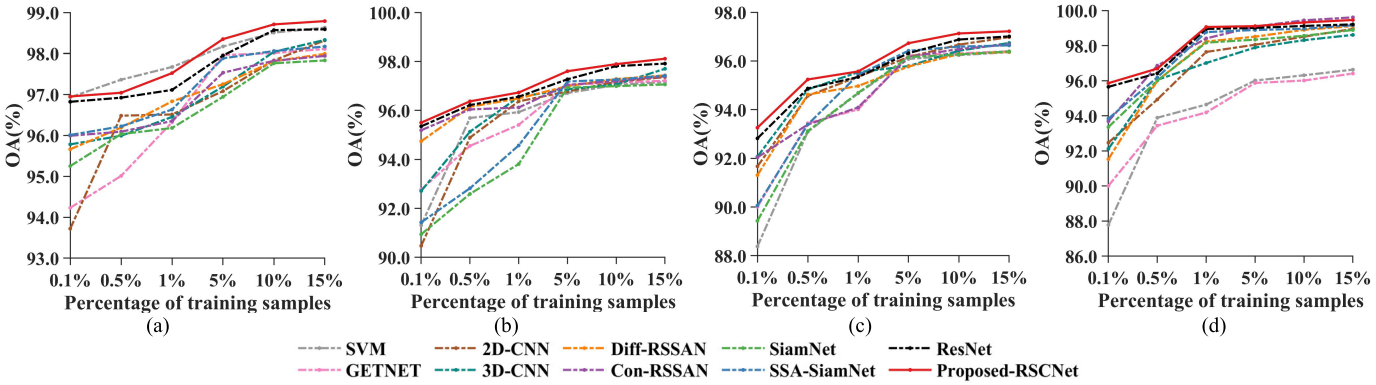


Fig. 4. Accuracy (in terms of OA) of different CD methods with different proportions of training samples for the: (a) Farmland dataset; (b) River dataset; (c) Hermiston dataset; and (d) Bay Area dataset.

4) Impact of the Number of Training Samples: In this section, we examined the effect of the number of training samples on different CD methods. A different number of training samples (0.1%, 0.5%, 1%, 5%, 10%, and 15%) were considered. As shown in Fig. 4, the OA increases when the proportion r ranges from 0.1% to 15% for the Farmland, River, and Hermiston datasets in all supervised methods. However, the increase in OA for the Bay Area dataset is much smaller when the proportion of training samples further increases from 1% to 15%. In general, the OA of the proposed RSCNet method is larger than that of ResNet and other benchmark methods. These results validate that the proposed RSCNet method can achieve comparable performance with fewer samples.

D. Comparison With Other Methods

To demonstrate the effectiveness of the proposed RSCNet method, we compared it with ten benchmark methods. They are: change vector analysis (CVA) [51], SVM [52], GETNET [19], 2-D-CNN [53], 3-D-CNN [54], Sub-residual spectral-spatial attention network (RSSAN) [32], Con-RSSAN [32], SiamNet [34], spectral–spatial-wise attention-based (SSA)-SiamNet [34], and ResNet. The aforementioned baseline from the modified ResNet was named ResNet for convenience. The result of subtracting bitemporal HSIs was classified in Sub-RSSAN. For SVM, 2-D-CNN, 3-D-CNN, Con-RSSAN, ResNet, and RSCNet, concatenate bitemporal HSIs were used as inputs to detect changes.

In the SVM-based method, the radial basis function kernel was adopted for all four datasets. In the benchmark methods, 2-D-CNN, 3-D-CNN, Sub-RSSAN, and Con-RSSAN were originally applied to the HSI classification problem with multiple land cover classes. Because the CD task is regarded as a binary classification in this study, if the above deep learning-based classification methods are applied directly to detect changes, overfitting can occur easily. Therefore, a dropout layer was added between the convolution layers for 2-D-CNN, 3-D-CNN, and the residual blocks for RSSAN. The dropout parameters were set to 0.2 for 2-D-CNN and 3-D-CNN, and 0.4 for Sub-RSSAN and Con-RSSAN. For the aforementioned deep learning-based benchmark methods developed originally for HSI classification, the L2 regularization coefficient was set to 0.001, and the binary cross entropy was used in loss function.

For a fair comparison, for each dataset, the numbers of pixel pairs in the training and test sets are the same for all supervised methods, which was set according to Table I.

1) Quantitative Comparison: The metrics describing the CD accuracy of each method (with the related standard deviations based on ten runs) for the four datasets are presented in Table II. From the analysis of the metrics, we can give four main observations.

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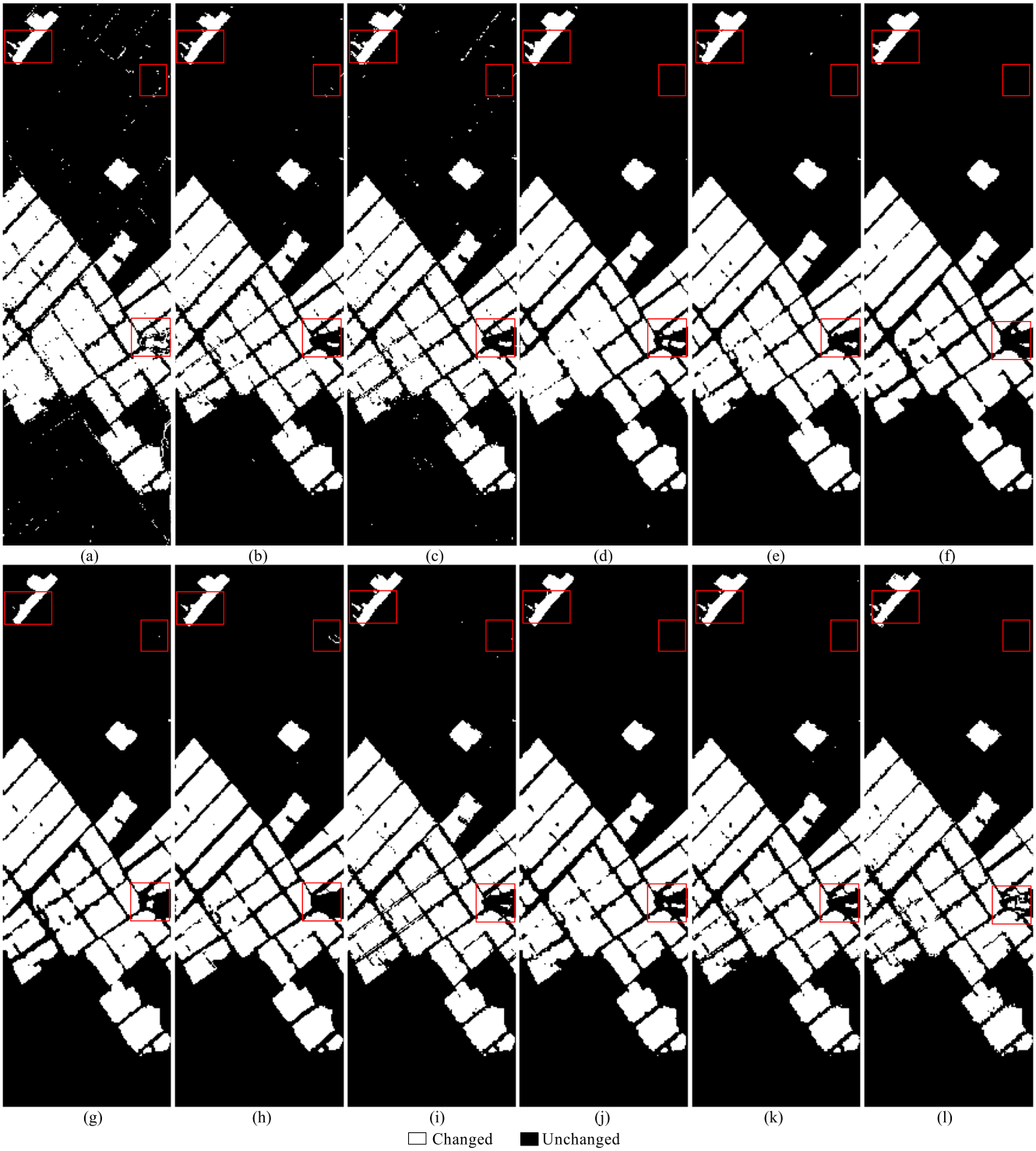


Fig. 5. CD maps of different methods for the Farmland dataset: (a) CVA; (b) SVM; (c) GETNET; (d) 2-D-CNN; (e) 3-D-CNN; (f) Sub-RSSAN; (g) Con-RSSAN; (h) SiamNet; (i) SSA-SiamNet; (j) ResNet; (k) proposed RSCNet; and (l) ground-reference map.

First, CVA obtained the lowest accuracy on all four datasets, which is especially prominent for the Bay Area dataset. This is because CVA is an unsupervised method without prior information. Thus, it cannot distinguish various changes accurately, especially when their features are very close to those of unchanged pixels.

Second, deep learning-based methods (i.e., GETNET, 2-D-CNN, 3-D-CNN, Sub-RSSAN, Con-RSSAN, SiamNet, SSA-SiamNet, ResNet, and RSCNet) generally produce greater accuracy than the SVM method. The reason is that compared with conventional machine learning-based methods, CNN-based methods can make fuller use of spatial information

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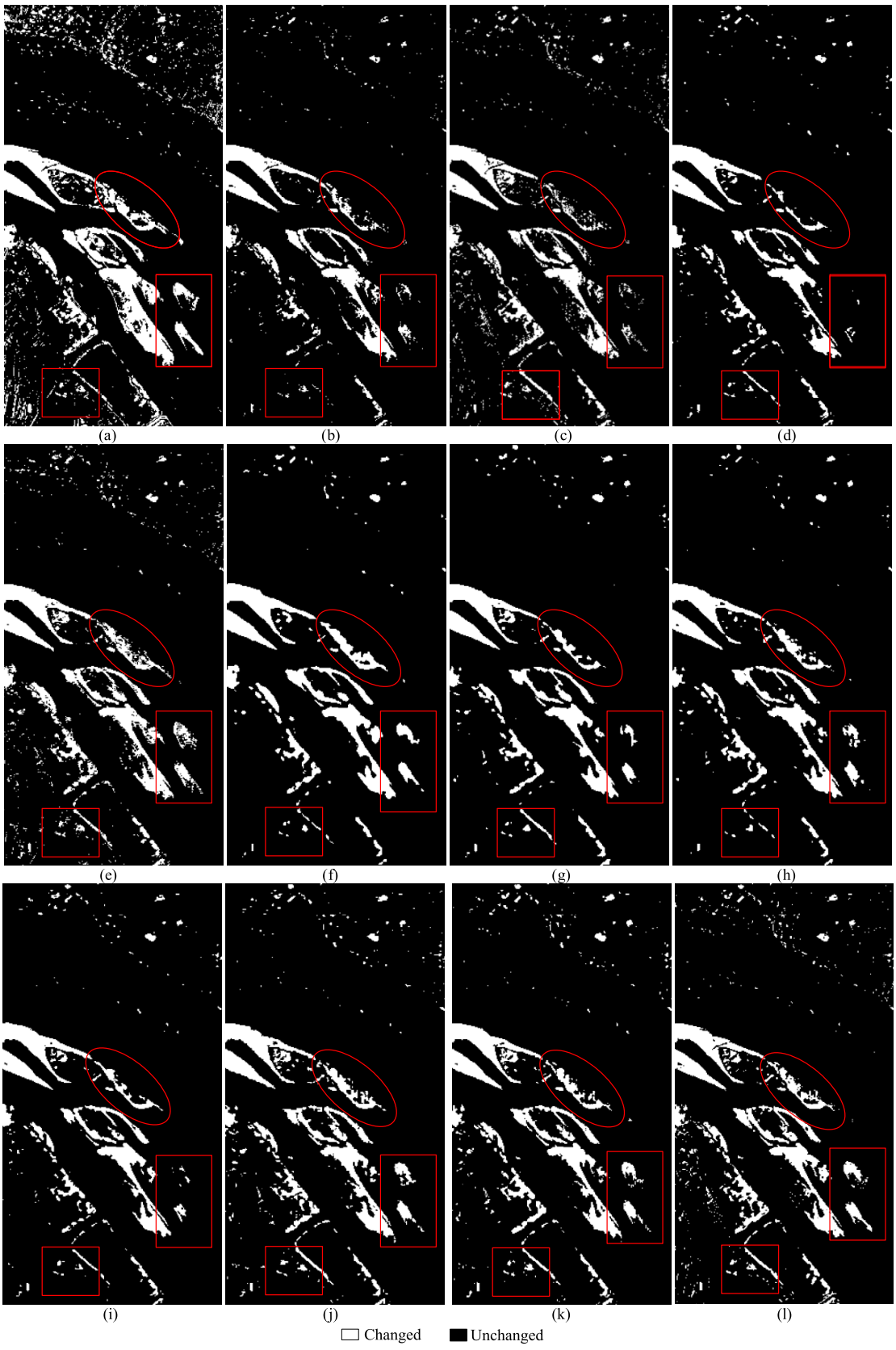


Fig. 6. CD maps of different methods for the River dataset: (a) CVA; (b) SVM; (c) GETNET; (d) 2-D-CNN; (e) 3-D-CNN; (f) Sub-RSSAN; (g) Con-RSSAN; (h) SiamNet; (i) SSA-SiamNet; (j) ResNet; (k) proposed RSCNet; and (l) ground-reference map.

and acquire more discriminative features through multilayer convolution operations, which enhances the CD performance.

Third, the results of the SSA-SiamNet are superior to those of SiamNet, as the former extracts spectral–spatial-wise

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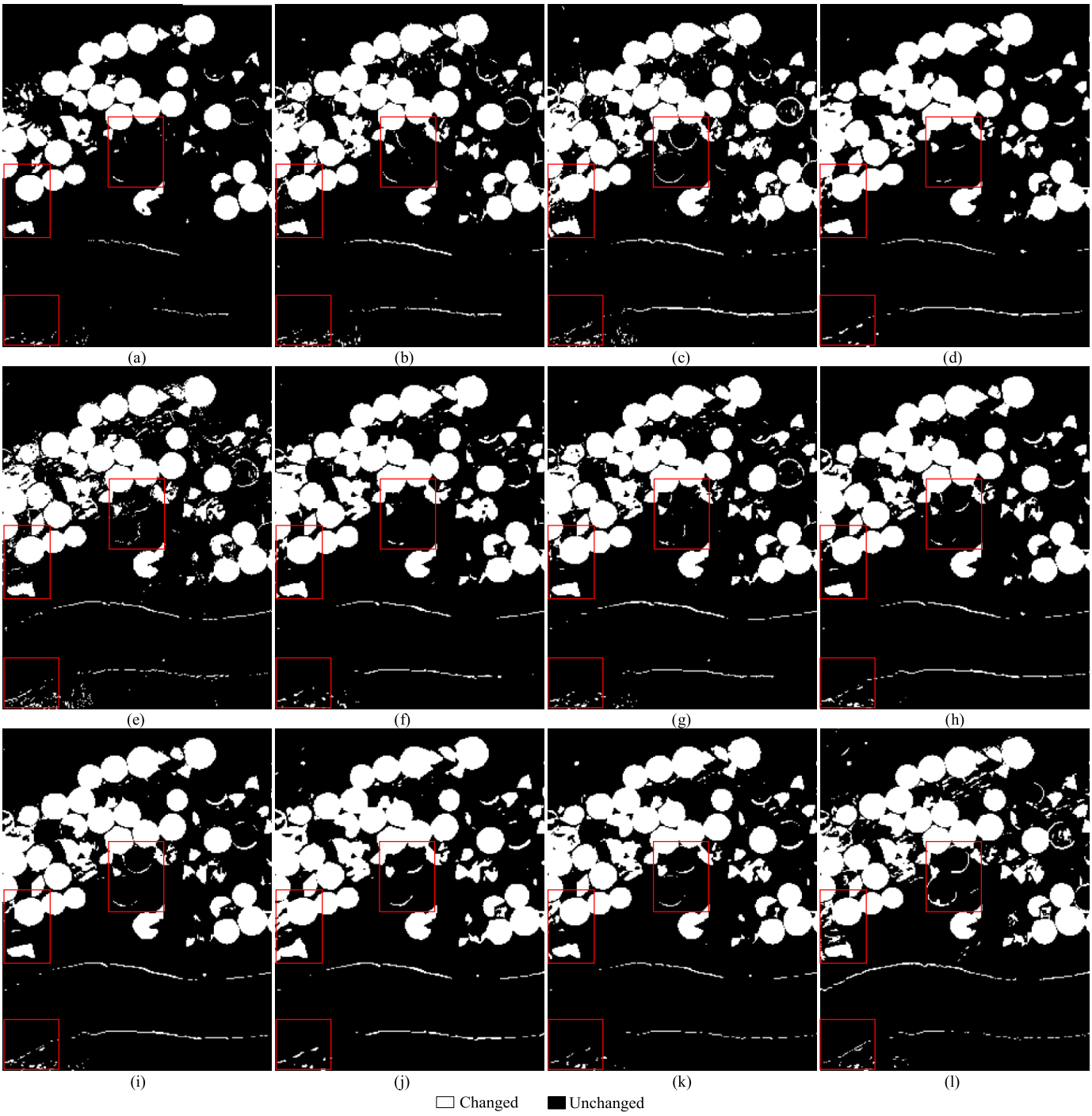


Fig. 7. CD maps of different methods for the Hermiston dataset: (a) CVA; (b) SVM; (c) GETNET; (d) 2-D-CNN; (e) 3-D-CNN; (f) Sub-RSSAN; (g) Con-RSSAN; (h) SiamNet; (i) SSA-SiamNet; (j) ResNet; (k) proposed RSCNet; and (l) ground-reference map.

features adaptively from the input patch pairs. For all four datasets, SSA-SiamNet and ResNet display similar detection accuracy and outperform the other deep learning-based methods, except the proposed RSCNet method. The findings indicate that the proposed revised ResNet method is more effective in dealing with the CD problem. In addition, Con-RSSAN achieved greater accuracy than Sub-RSSAN for all datasets.

Finally, it is clear that the most accurate results are produced by the proposed RSCNet method. More precisely, the increase in the OA of the RSCNet over ResNet for the Farmland, River, Hermiston, and Bay Area datasets was 0.40%, 0.33%, 0.41%, and 0.11%, respectively.

2) Qualitative Comparison: Ground-reference maps for the four datasets are shown in (l) of Figs. 5–8, where the black, white, and gray parts represent the unchanged, changed, and unknown pixels, respectively. The CD maps produced by the 11 methods are shown in Figs. 5–8. The areas with the major differences with maps are pointed out by red squares and circles. Overall, the CVA method shows a more apparent presence of noise than the other supervised methods, illus-

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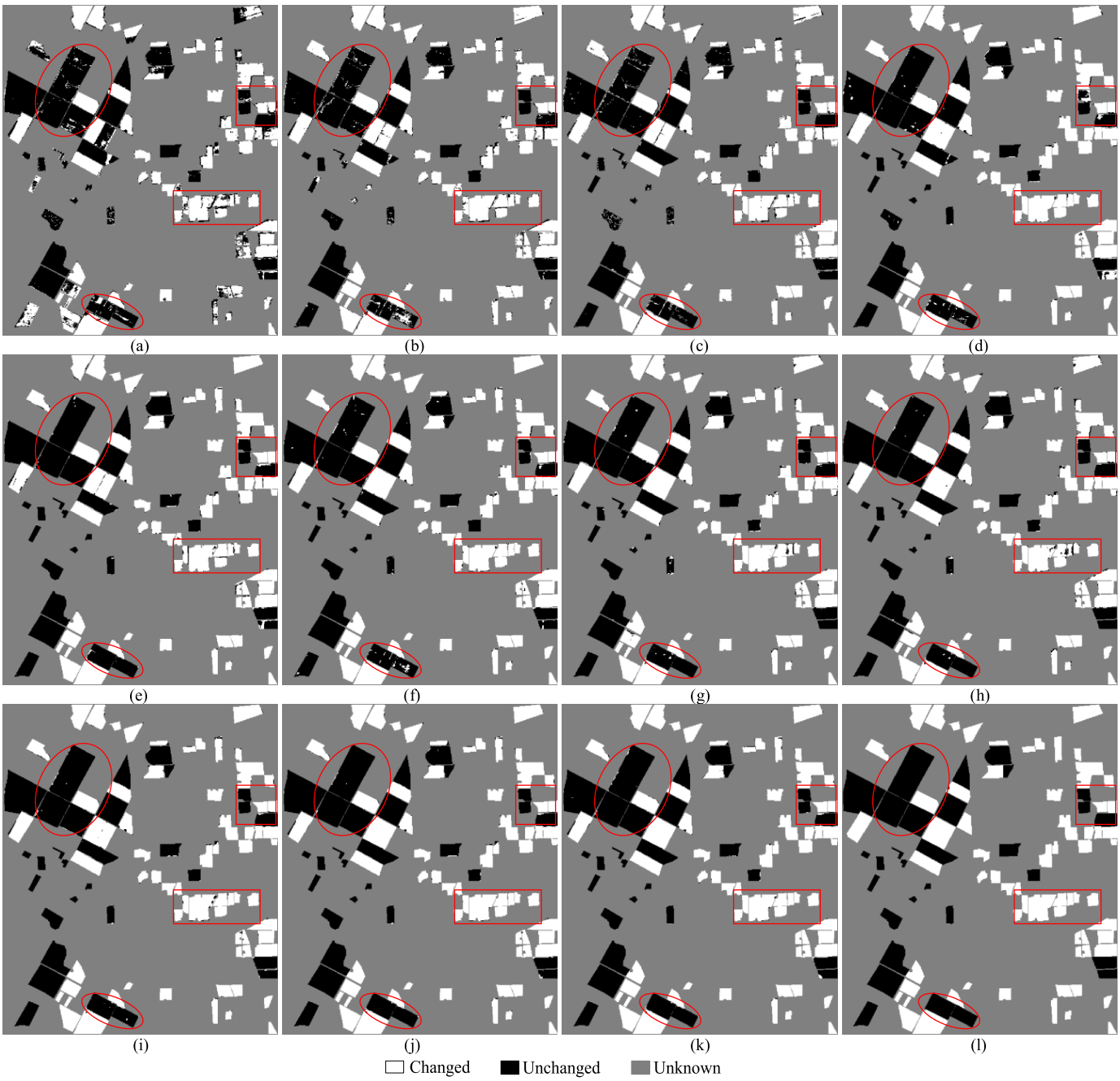


Fig. 8. CD maps of different methods for the Bay Area dataset: (a) CVA; (b) SVM; (c) GETNET; (d) 2-D-CNN; (e) 3-D-CNN; (f) Sub-RSSAN; (g) Con-RSSAN; (h) SiamNet; (i) SSA-SiamNet; (j) ResNet; (k) proposed RSCNet; and (l) ground-reference map.

trating the importance of using training information in CD. In general, compared with the SVM method, the CD maps of the deep learning-based methods are closer to the ground-reference map, which is consistent with the quantitative results in Table II. Furthermore, the CD map of the proposed RSCNet method is closer to the ground-reference map on each dataset.

3) Computational Cost: Table III lists the model complexity of the nine deep learning-based methods in terms of computational costs and number of trainable parameters. The computational cost was calculated by floating-point operations (FLOPs). It is evident that GETNET requires the largest number of parameters and FLOPs among the nine methods for all datasets. This is due to the relatively large patch of size $b \times b \times 1$ and the large number of parameters to be determined. The number of parameters and FLOPs of the ResNet and the proposed RSCNet are approximately similar and far less than those of other methods on all datasets. Furthermore, the proposed RSCNet involves the lowest FLOPs among the nine methods for the four datasets. The training time per sample for different methods suggests that the RSCNet and ResNet require less computation time compared to other methods, thus confirming the superiority of the proposed method.

E. Ablation Analysis

To further demonstrate the validity of different modules used in the proposed method, ablation experiments were

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TABLE II

ACCURACY OF DIFFERENT CD METHODS ON THE FOUR DATASETS (THE BOLD VALUES INDICATE THE MOST ACCURATE FOR EACH VARIABLE AND EACH DATASET)

Data set	Method	Metrics				
		OA (%)	Kappa×100	F1 (%)	Pr (%)	Re (%)
Farmland	CVA	96.08	90.63	93.42	91.01	95.97
	SVM	98.17	95.55	96.84	97.05	96.62
	GETNET	97.96±1.18	95.07±2.60	96.51±1.87	95.64±2.08	**97.40±1.56**
	2D-CNN	97.08±0.31	92.92±0.71	94.98±0.50	94.75±1.62	95.26±1.48
	3D-CNN	97.19±0.28	93.19±0.65	95.17±0.46	94.87±1.58	95.53±1.53
	Sub-RSSAN	97.25±0.14	93.28±0.34	95.21±0.24	95.98±0.66	94.46±0.62
	Con-RSSAN	97.53±0.10	93.99±0.25	95.73±0.18	95.74±0.15	95.72±0.36
	SiamNet	96.94±0.21	92.53±0.23	94.67±0.22	95.71±0.35	93.66±0.42
	SSA-SiamNet	97.88±0.12	94.85±0.28	96.34±0.20	96.24±0.36	96.44±0.29
	ResNet	97.95±0.20	95.03±0.47	96.48±0.33	96.22±0.95	96.73±0.92
	Proposed RSCNet	**98.35±0.34**	**95.98±0.08**	**97.15±0.06**	**97.33±0.21**	96.96±0.22
River	CVA	94.22	69.72	72.84	61.56	89.16
	SVM	96.71	76.93	78.68	90.17	69.79
	GETNET	97.04±0.23	80.57±1.35	82.18±1.23	85.64±2.45	78.98±2.38
	2D-CNN	96.77±0.13	78.89±1.20	80.71±1.15	81.51±2.91	80.22±2.32
	3D-CNN	96.86±0.12	80.52±1.19	82.24±1.14	80.94±1.31	83.58±2.70
	Sub-RSSAN	96.99±0.06	80.25±0.26	81.89±0.22	85.81±0.94	78.31±0.67
	Con-RSSAN	97.04±0.18	79.64±0.35	81.23±0.28	90.24±0.72	73.86±0.54
	SiamNet	96.93±0.45	79.59±0.23	81.25±0.20	86.67±0.51	76.47±0.32
	SSA-SiamNet	97.18±0.32	80.53±0.19	82.04±0.21	**91.89±0.85**	74.10±0.43
	ResNet	97.27±0.08	82.46±0.47	83.95±0.42	85.74±0.72	82.23±0.59
	Proposed RSCNet	**97.60±0.06**	**84.62±0.74**	**85.93±0.72**	87.50±1.69	**84.41±0.26**
Hermiston	CVA	93.87	80.85	84.59	97.61	74.63
	SVM	96.08	88.49	91.00	94.17	88.03
	GETNET	96.23±0.07	89.09±0.34	91.50±0.32	92.99±1.84	90.16±2.25
	2D-CNN	96.16±0.51	88.93±1.45	91.40±1.12	92.18±1.32	90.64±1.07
	3D-CNN	95.79±0.10	87.78±0.35	90.48±0.29	92.17±0.43	88.87±0.89
	Sub-RSSAN	95.77±0.11	87.81±0.32	90.53±0.25	91.37±0.46	89.71±0.52
	Con-RSSAN	96.17±0.25	88.94±0.72	91.40±0.56	92.56±0.68	90.28±0.57
	SiamNet	96.16±0.32	88.86±0.82	91.31±0.61	93.41±1.84	89.35±0.73
	SSA-SiamNet	96.42±0.45	89.52±0.17	91.80±0.14	**94.88±0.29**	88.93±0.53
	ResNet	96.32±0.04	89.28±0.20	91.64±0.18	93.75±1.00	89.63±1.21
	Proposed RSCNet	**96.73±0.05**	**90.53±0.15**	**92.63±0.12**	93.98±0.50	**91.32±0.05**
Bay Area	CVA	86.83	73.78	86.83	93.20	81.28
	SVM	94.64	89.26	94.93	96.11	93.77
	GETNET	95.87±0.63	91.71±1.29	96.09±0.49	97.07±1.29	95.14±1.38
	2D-CNN	97.64±0.43	95.26±0.87	97.78±0.42	98.16±0.65	97.41±0.87
	3D-CNN	97.89±0.48	95.75±0.95	98.01±0.45	98.27±0.58	97.77±0.83
	Sub-RSSAN	98.23±0.50	96.46±1.01	98.35±0.47	98.53±0.50	98.17±0.64
	Con-RSSAN	98.41±0.47	96.80±0.95	98.50±0.44	98.64±0.49	98.38±0.57
	SiamNet	98.17±0.79	96.33±0.97	98.27±0.51	**99.28±0.43**	97.29±0.69
	SSA-SiamNet	98.77±0.46	97.53±0.79	98.85±0.41	99.10±0.59	98.59±0.72
	ResNet	98.95±0.10	97.90±0.20	99.02±0.09	99.12±0.12	98.92±0.18
	Proposed RSCNet	**99.06±0.04**	**98.11±0.08**	**99.12±0.04**	99.17±0.18	**99.07±0.21**

performed while the other parameter settings were fixed according to Section IV-B.

1) Analysis of SCConv and Vanilla Convolution: To further investigate the effectiveness of the proposed SCConv compared to the vanilla convolution, we comprehensively analyzed the ResNet and the proposed RSCNet sections. The results are presented in Tables II and III and in Fig. 4. In general, the results of the proposed RSCNet are more accurate than those of ResNet for all datasets. This indicates that compared with the vanilla convolution, the networks in the proposed SCConv can generate richer and more discriminative feature representations.

2) Analysis of the Interpolation and Deconvolution Strategies: To restore the spatial size of features, a bicubic interpolation is applied in the proposed RSCNet. The deconvolution layer is also used to realize the nonlinear up-sampling. We compared the bicubic interpolation and deconvolution strategies based on RSCNet, and the results are shown in Fig. 9 and Table IV. Overall, the accuracy of bicubic interpolation and deconvolution is similar, but the bicubic interpolation involves less computational costs.

3) Analysis of Different Interpolation Strategies: Different interpolation strategies, including bilinear, nearest neighbor, and bicubic, are considered to further explore the influence of the proposed SCConv. As shown in Table V (the results are all based on average pooling), the difference in the accuracy of different interpolation strategies is not obvious in the four

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TABLE III

COMPUTATIONAL COSTS OF DIFFERENT METHODS FOR THE FOUR DATASETS (THE BOLD VALUES INDICATE THE LEAST COMPUTATIONAL COST RESULT FOR EACH VARIABLE AND EACH DATASET)

Dataset	Method	Parameters (K)	FLOPs (M)	Computation time (ms/sample)
Farmland	GETNET	93420	55080	113.42
	2D-CNN	73.13	104.24	51.96
	3D-CNN	155.07	1130	126.21
	Diff-RSSAN	49.25	41.25	57.55
	Con-RSSAN	103.05	83.63	70.23
	SiamNet	87.84	39.36	68.23
	SSA-SiamNet	88.60	39.60	101.16
	ResNet	**34.47**	18.92	**31.34**
	Proposed RSCNet	36.99	**17.82**	46.71
River	GETNET	154180	90910	179.79
	2D-CNN	91.71	132.77	53.32
	3D-CNN	159.19	1470	120.97
	Diff-RSSAN	63.65	52.87	57.15
	Con-RSSAN	135.14	107.72	70.65
	SiamNet	106.42	134.15	77.34
	SSA-SiamNet	107.18	134.80	104.53
	ResNet	**37.22**	20.50	**28.24**
	Proposed RSCNet	39.75	**19.41**	46.37
Hermiston	GETNET	92180	54350	210.70
	2D-CNN	99.49	103.57	50.40
	3D-CNN	72.07	1120	112.30
	Diff-RSSAN	37.17	40.99	66.24
	Con-RSSAN	79.53	83.09	104.38
	SiamNet	110.15	104.95	87.36
	SSA-SiamNet	111.52	105.60	119.03
	ResNet	**34.40**	18.88	**48.07**
	Proposed RSCNet	36.93	**17.79**	72.08
Bay Area	GETNET	198270	116910	230.71
	2D-CNN	139.81	150.02	68.79
	3D-CNN	277.03	1670	144.50
	Diff-RSSAN	72.68	59.98	70.72
	Con-RSSAN	156.25	122.72	84.90
	SiamNet	166.08	151.40	97.66
	SSA-SiamNet	167.32	152.05	142.33
	ResNet	**13.55**	67.65	**33.13**
	Proposed RSCNet	14.31	**39.77**	46.77

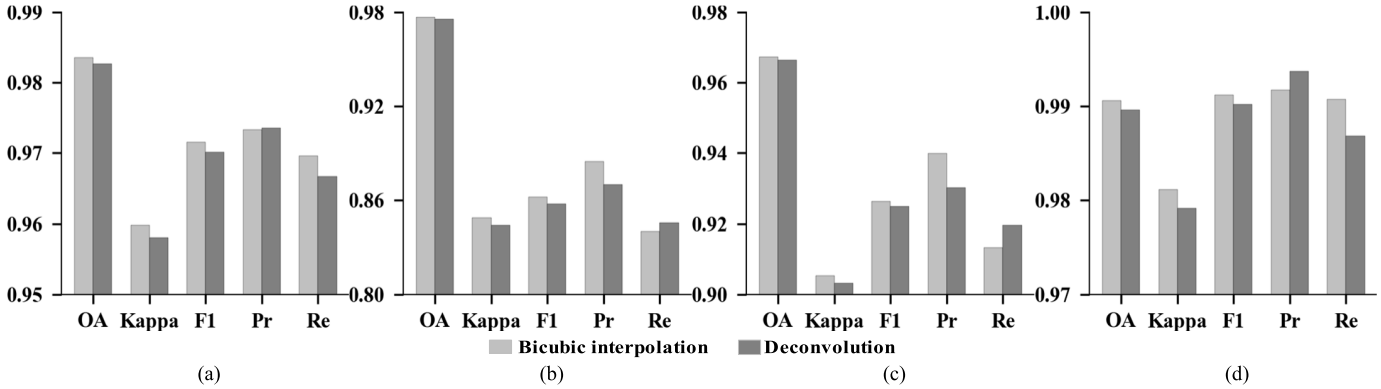


Fig. 9. Accuracy of the proposed RSCNet with different up-sampling strategies for the: (a) Farmland dataset; (b) River dataset; (c) Hermiston dataset; and (d) Bay Area dataset.

datasets, indicating the robustness of the proposed RSCNet to the algebraic interpolation strategy.

4) Analysis of Average Pooling and Max Pooling: We also investigate the impact of different pooling types on the performance of the proposed method. The average pooling and max pooling operators are placed separately in SCConv, with all other configurations kept fixed. The average pooling operator leads to greater accuracy than the max pooling operator for all datasets, except for the River dataset (Fig. 10). This may be because the average pooling operator produces connections among locations within the whole pooling window, which can capture more local contextual information. For a fair comparison, the pooling type in SCConv of the proposed RSCNet is fixed as the average pooling operator.

5) Analysis of Different Input Modes: The input mode for bitemporal HSIs can affect the CD results. We analyzed the

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TABLE IV

COMPUTATIONAL COSTS OF THE PROPOSED RSCNET WITH DIFFERENT UP-SAMPLING STRATEGIES

Up-sampling strategies	Farmland		River		Hermiston		Bay Area	
	Parameters (K)	FLOPs (M)	Parameters (K)	FLOPs (M)	Parameters (K)	FLOPs (M)	Parameters (K)	FLOPs (M)
Bicubic Interpolation	36.99	17.82	39.75	19.41	36.93	17.79	14.31	39.77
Deconvolution	41.63	20.49	44.39	22.07	41.57	20.45	17.52	50.03

TABLE V

RESULTS PRODUCED BY THE PROPOSED RSCNET WITH DIFFERENT INTERPOLATION STRATEGIES (THE BOLD VALUES INDICATE THE MOST ACCURATE RESULT FOR EACH VARIABLE AND EACH DATASET)

Interpolation strategies	Farmland			River			Hermiston			Bay Area		
	OA (%)	Kappa ×100	F1 (%)	OA (%)	Kappa ×100	F1 (%)	OA (%)	Kappa ×100	F1 (%)	OA (%)	Kappa ×100	F1 (%)
Bilinear	**98.35**	**95.98**	**97.15**	**97.60**	84.62	85.93	96.67	90.38	92.52	98.96	97.91	99.02
Nearest Neighbor	98.32	95.91	97.09	97.57	84.31	85.64	**96.73**	**90.53**	**92.63**	**99.06**	**98.11**	**99.12**
Bicubic	98.28	95.83	97.04	97.58	**84.64**	**85.96**	96.60	90.15	92.34	98.64	97.27	98.71

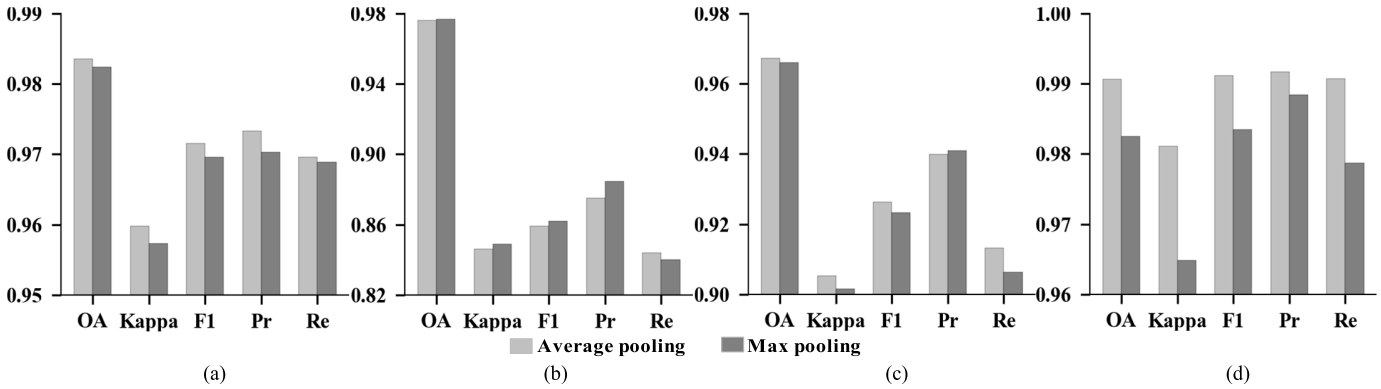


Fig. 10. Accuracy of the proposed RSCNet under average pooling and max pooling operations for the: (a) Farmland dataset; (b) River dataset; (c) Hermiston dataset; and (d) Bay Area dataset.

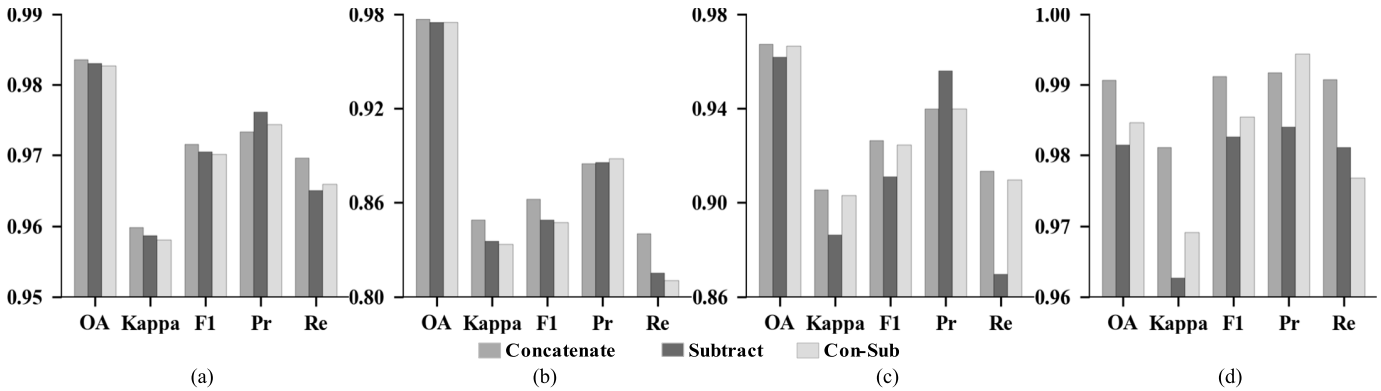


Fig. 11. Accuracy of the proposed RSCNet under different input modes for the: (a) Farmland dataset; (b) River dataset; (c) Hermiston dataset; and (d) Bay Area dataset.

performances of different input modes based on the proposed RSCNet. The input modes include the subtraction I_d and concatenation I_s of bitemporal HSIs and the concatenation of them I_c (named as Con–Sub and $I_c = [I_s, I_d]$). From Fig. 11, it is seen that the results of I_s are superior to those of I_d and I_c for the four datasets. The difference between the three modes is especially obvious for the Bay Area dataset. The reason is that although the subtraction operation emphasizes the differences between bitemporal maps, it leads to obvious information loss of the bitemporal HSIs. Moreover, the concatenation operation retains all original information of both HSIs. However, the Con–Sub input mode with many redundant channels requires more computational resources and might cause overfitting.

V. DISCUSSION

The advantage in terms of CD accuracy of the proposed RSCNet method over the benchmark methods is clear from the experiment and is mainly due to the use of the RSC

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module that is composed of the proposed SCConv and the residual block. The proposed RSC module can adaptively build the long-range spatial and interspectral dependencies around each spatial location, which is helpful to emphasize the informative channels and locations and to suppress the less informative ones to refine the spectral–spatial features adaptively. Differently from existing CNN-based methods in HSI processing tasks, the proposed RSCNet method can generate more discriminative representations by introducing only few extra parameters and a reduced complexity. Moreover, the proposed method achieves a satisfactory balance between CD accuracy and computational cost.

The recent remarkable progress of CNN-based methods in HSI processing tasks mainly focused on tuning network architectures and adding additional hand-designed blocks, which involve too much manual human activity and require additional parameters in the fitting process. On the contrary, the designed SCConv is complementary to attention and is an efficient way to generate more expressive feature representations by internally augmenting the basic convolutional transformation per layer. This is achieved with no additional parameters. In general, grouped convolutions homogeneously conduct individual convolutional transformations in multiple parallel branches [55] or in a hierarchical way [56]. For the SCConv, however, different portions of convolutional filters can be exploited in a heterogeneous way. Therefore, the SCConv can fuse information from two different spatial scale spaces, which results in more discriminative feature representations. In addition, the RSCNet method is fairly robust to hyperparameters, such as kernel number, batch size, and patch size. This property is helpful to promote applicability in practice under various conditions.

Differently from the vanilla convolution where the input data are processed in the original space homogeneously, the designed SCConv first transforms the input to low-dimensional embedding through a down-sampling strategy. Then, the low-dimensional embedding transformed by filter portion is adopted to calibrate the convolutional transformations of the filters within another portion. In this study, the effectiveness of RSCNet has been demonstrated through experiments based on HSIs acquired by the EO-1 satellite. However, considering the limited spatial resolution of approximately 30 m, subpixel information derived from the image super-resolution process may assist in detecting changes for HSIs. For this reason, in future research, we plan to design the convolution with the aid of subpixel information. Especially, the fine spatial resolution embedding, transformed from the input in the original space through the super-resolution CNN strategy, can be used to calibrate the convolutional transformations. In this case, the prospective convolution with the aid of subpixel information seems potentially suitable for the CD task between multiresolution images. It is reasonable to transform the two images to the same spatial resolution by tuning the corresponding convolution parameters.

The proposed method is also potentially applicable to other remote sensing images, such as MSI, very-high-resolution (VHR) images, and synthetic aperture radar (SAR) images. In these cases, the hyperparameters, i.e., the kernel number, batch size, and patch size, should be properly determined. For example, for VHR images with more abundant spatial information, it is rational to increase the patch size to make fuller use of the spatial texture. Unlike HSIs, MSIs contain few spectral bands. Thus, the numbers of kernels may be reduced. On the other hand, the proposed SCConv can be applied to other HSI tasks based on the deep learning framework, such as land cover mapping, target detection, anomaly detection, and image super-resolution. Meanwhile, the proposed method can be used in several practical applications, such as forest and farmland mapping, ecological and environmental change monitoring, and urban development monitoring.

There are also several limitations of the proposed method. First, RSCNet deals with binary CD to distinguish between "changed" or "no changed" pixels, rather than multiple changes that solve "from–to" problems. Second, the proposed method is supervised, and thus, its performance may be limited by the lack of available ground-reference labels for the changed and unchanged classes. To address this issue, it is worthwhile to further explore an automatic solution to create more training samples based on the limited ones collected. Accordingly, an unsupervised method with calibration property for multiple CD will be explored in further research.

VI. Conclusion

In this study, an end-to-end RSCNet deep learning framework has been proposed to detect land cover changes in bitemporal HSIs. The proposed method uses a modified ResNet designed specifically for HSI CD as a baseline to learn spectral–spatial features from the input patch pairs. A novel SCConv that can heterogeneously exploit convolutional filters nested in the convolutional layer is introduced to generate more discriminative representations. Then, the designed RSC module, which integrates the introduced SCConv and the residual block, is stacked in the proposed framework to adaptively build interspatial and interspectral dependencies around each spatial location by introducing very few parameters. Finally, it is worth working that the proposed RSC module can be easily extended to other HSI processing tasks. The proposed RSCNet method was examined using four real HSI datasets and was compared with ten state-of-the-art methods. It was shown that RSCNet produced consistently more accurate results.

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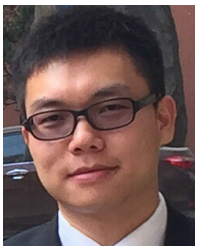
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